

Foliar Biochemistry Modeling using Non-Invasive Hyperspectral Methods

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Abstract—Hyperspectral techniques provides a non-invasive and efficient method towards estimating foliar nutrient content in crops, aiding precision agriculture. We applied a stepwise linear regression to model nitrogen (N) and potassium (K) levels in crop samples using laboratory spectra (350–2500 nm) in their raw, first derivative, and continuum-removed forms. Model performance was evaluated at three alpha-to-enter levels ($\alpha = 0.005, 0.01$, and 0.05). The raw spectra achieved best performance at $\alpha = 0.05$ (N: $S = 0.13$, adj. $R^2 = 48.86\%$; K: $S = 0.15$, adj. $R^2 = 22.21\%$), the first derivative at $\alpha = 0.01$ (N: $S = 0.04$, adj. $R^2 = 96.09\%$; K: $S = 0.10$, adj. $R^2 = 64.75\%$), and the continuum-removed spectra at $\alpha = 0.05$ (N: $S = 0.09$, adj. $R^2 = 78.18\%$; K: $S = 0.13$, adj. $R^2 = 39.95\%$). This outcome also revealed that different spectral transformations perform optimally under different conditions. Key spectral regions for N and K prediction varied based on the spectral transformation applied. For raw spectra, the most informative wavelengths were found in the UV, near-infrared (NIR), and shortwave infrared (SWIR) regions, particularly around 370–380 nm, 1450 nm, and 2270 nm for N, and 710 nm, 1790 nm, and 2290 nm for K. The first derivative transformation emphasized finer spectral details, with key wavelengths for N prediction observed in the visible-NIR (380–670 nm, 800–920 nm) and longer SWIR bands (1260–2350 nm), while K prediction relied on 680–940 nm and 1700–2260 nm. The continuum-removed transformation highlighted absorption features, with significant wavelengths for N around 620–690 nm, 1030–1450 nm, and 1540–2060 nm, and for K in the visible and SWIR regions (660–1290 nm). These findings were consistent with previous studies, showing that the identified spectral bands correspond to important physiological and biochemical absorption features in vegetation. This highlights the importance of spectral transformations and wavelength selection in optimizing hyperspectral data for nutrient monitoring in crops for application in precision agriculture.

Index Terms—Precision agriculture, Foliar biochemistry, Hyperspectral data, Stepwise linear regression.

I. INTRODUCTION

FOLIAR biochemical composition like chlorophyll, nitrogen, phosphorus, lignin, cellulose and water are key variables in quantifying and detecting changes in ecosystem processes like carbon sequestration, net primary productivity, nutrient cycling and energy exchange within the ecosystem [1]–[3]. Traditional methods of estimating foliar biochemical content involve *destructive sampling* followed by *lab-based chemical analysis*, which, while accurate, are laborious, expensive, and unfeasible for large-scale applications in precision agriculture. However, *non-invasive hyperspectral spectroscopy* provides a rapid and efficient alternative by analyzing leaf spectra to estimate plant biochemical content [4], [5]. By leveraging spectral bands correlated with biochemical absorption

features, hyperspectral methods can estimate foliar properties in a non-invasive way.

Several methods for such biochemical estimation have been examined. An investigation by *Aber et al. (1994)* [6] showed that combination of linear mixing models and end-member analysis is not a suitable approach for accurately quantifying the concentrations of major foliar constituents. Studies of *Johnson et al. (1994)* [7], *LaCapra et al. (1996)* [8] and *Martin and Aber (1997)* [9] showed a correlation of plant canopy chemistry to spectrometer data. Continuing to the development of such studies, *Curran (1989)* [5] and *Martin and Aber (1997)* [9], suggested using empirical methods to estimate canopy chemical constituents based on remote spectroscopic measurements. These studies were based on using *stepwise multiple linear regression* to predict canopy chemical composition using derivative reflectance spectra. However the regression model using raw and derivative leaf spectra were found to be inconsistent at leaf and canopy scales in these studies. *Kokaly and Clark (1999)* [4] applied *continuum removal technique* [10] to broad absorption features to determine normalize band depths in dry leaf spectra to determine Nitrogen, Cellulose and Lignin concentration using stepwise linear regression. They were able to dissect certain wavelength bands with high correlation to leaf chemistry.

Building upon these past studies, we aim to evaluate the effectiveness of *raw spectra*, *first derivative spectra*, and *continuum-removed spectra* in estimating crop foliar N and K content using a stepwise linear regression model. By systematically varying the feature inclusion criterion, we seek to optimize predictor selection and enhance model performance in order to achieve our objectives- 1) To assess the predictive accuracy of different spectral preprocessing techniques in estimating foliar nitrogen and potassium concentrations and 2) To identify key wavelength regions that contribute most to N and K estimation across spectral transformations. With this approach, we look forward to improve the robustness of hyperspectral biochemical estimation models, contributing to efficient non-invasive foliar nutrient analysis.

II. DATA AND METHODS

A dataset consisting of 92 spectra ranging from 350 nm to 2500 nm at 1 nm spectral sampling, for 92 different crop samples collected in controlled laboratory settings along with the corresponding nutrient (N and K) measurement was used for this analysis. Raw spectrum of a sample is shown in Figure 1. The distinct vegetation features like *green reflectance*,

sharp red-edge, and water absorptions (1400 nm, 1900 nm) can be observed clearly in the spectrum. We utilized such lab collected spectra to model N and K using linear regression model given as:

$$Y = \beta_0 + \sum_{i=1}^p \beta_i X_i + \epsilon \quad (1)$$

here Y represents the predicted nutrient content (N or K), X_i are the predictor variables (spectral bands in this case), β_0 is the intercept, β_i are the regression coefficients, p is the number of predictor variables, and ϵ is the error term. Since there are almost 2200 spectral bands in each sample but only 92 ground-truth observations, directly fitting a linear regression model using all available wavelengths is not feasible. The number of predictors far exceeds the number of samples, leading to an underdetermined system where a unique solution cannot be obtained. Furthermore, many adjacent spectral bands exhibit *high collinearity*, meaning they present redundant information in the spectrum. This can make it difficult for model to interpret which wavelengths are truly important for nutrient prediction. To overcome these challenges, a *stepwise linear regression* approach with an alpha-to-enter (α) criterion was applied to systematically select the most relevant wavelengths. This method incrementally adds predictors based on statistical significance, ensuring that only the most informative spectral bands contribute to the model. If α is very small then only the strongest predictors are selected for the model and vice-versa. By tuning the α threshold, the model balances complexity and predictive power, allowing for a more robust and interpretable selection of spectral features relevant to N and K estimation.

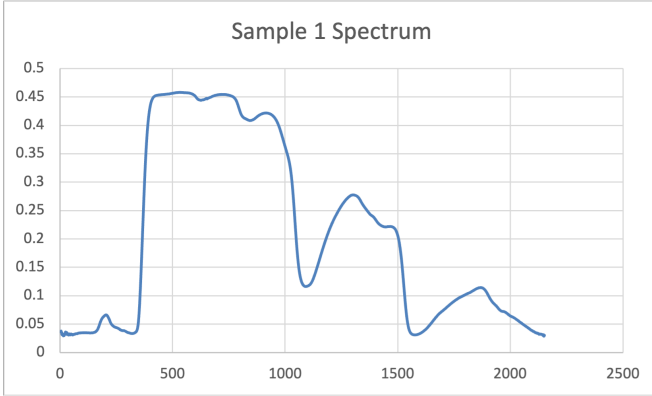
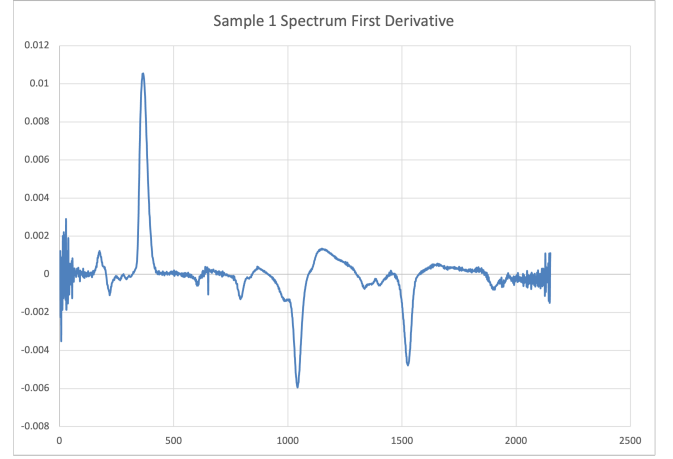
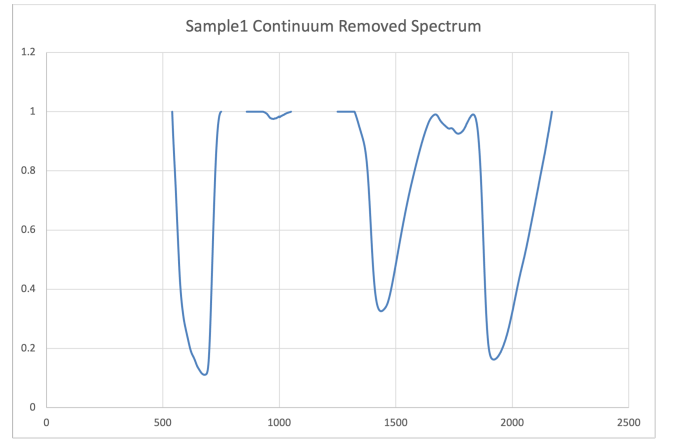


Fig. 1: Sample of raw spectrum collected in the lab
Note: The plot is spectral samples vs reflectance value.

Additionally, two other preprocessed versions of the raw spectra were used for modeling- (1) First Derivative Spectra and (2) Continuum-Removed Spectra. This was to assess whether preprocessing improves the predictive capability of the model. The sample of the first derivative spectrum and continuum-removed spectrum for the same sample is shown in Figure 2. The *first derivative transformation* amplifies spectral features by underscoring changes and highlighting subtle absorption features. Mathematically, the first derivative of the reflectance spectrum $R(\lambda)$ is computed as:



(a)



(b)

Fig. 2: Preprocessed Spectrum: (a) First derivative spectrum and (b) Continuum removed spectrum.

$$R'(\lambda) = \frac{R(\lambda + \Delta\lambda) - R(\lambda)}{\Delta\lambda} \quad (2)$$

where $R'(\lambda)$ represents the first derivative reflectance, $R(\lambda)$ is the original reflectance at wavelength λ , and $\Delta\lambda$ is the spectral sampling interval (1 nm in our case).

Similarly, the *continuum removal transformation* isolates absorption features by fitting a local linear continuum over specific wavelength ranges and normalizing the spectra accordingly. In our case, the continuum was calculated separately over three key spectral regions: (1) 540–750 nm (red absorption and red-edge feature), (2) 860–1050 nm (potential N/K absorption features), and (3) 1250–2170 nm (additional potential N/K absorption features). For each region, the *continuum* was estimated using a linear function:

$$R_{\text{continuum}}(\lambda) = m\lambda + b \quad (3)$$

where m is the slope and b is the intercept, both computed from the reflectance values at the endpoints of the selected wavelength range. The *continuum-removed spectrum* was then calculated as:

$$R_c(\lambda) = \frac{R(\lambda)}{R_{\text{continuum}}(\lambda)} \quad (4)$$

This method ensures that absorption features are normalized relative to their local spectral background rather than the full spectrum. By applying this approach, variations in absorption depth across different regions are more effectively captured, which strengthens the contribution of absorption features in nutrient modeling. These preprocessing methods were applied to enhance the predictive capability of the stepwise regression model by improving feature selection and separability in the spectral data.

III. RESULTS

The results for Nitrogen (N) and Potassium (K) modeling using- 1) Raw spectra, 2) First derivative spectra and 3) Continuum removed spectra for 3 α -levels: 0.005 (strict selection), 0.01 (mid-value), 0.05 (standard value) is shown in Table I. The best-performing models for each preprocessing technique and nutrient are highlighted in *light green* color. It can be noted that the *first derivative spectra* yielded the highest predictive performance for both N and K, notably at $\alpha = 0.01$, where the adjusted R^2 values were 96.09% and 64.75%, with corresponding 5-fold R^2 values of 94.20% and 60.94%, respectively. For the *raw spectra*, the best results were obtained at $\alpha = 0.05$, where the adjusted R^2 for N was 48.86% and for K was 22.21%, indicating lower predictive power compared to the first derivative spectra. The *continuum removal* approach showed intermediate performance, with the best results also at $\alpha = 0.05$, yielding an adjusted R^2 of 78.18% for N and 39.95% for K.

Apparently, the first derivative spectra did not yield results at $\alpha = 0.05$. This is because at this level too many predictor bands were selected, leading to overfitting of the model, which subsequently failed to generate any outcome. This shows the importance of tuning α -level to balance predictive power and model complexity. Overall, the first derivative transformation significantly improved predictive accuracy, particularly for N estimation, while the continuum removal method provided moderate improvement as compared to using raw spectra.

This highlights the importance of suitable preprocessing for improved modeling of nutrient content.

Also we assessed the wavelength bands that contributed most to the best prediction outcome. The list of such bands for both N and K prediction is presented in Table II. From the table, it can be seen that the selected wavelength regions vary depending on the preprocessing method, and many of these correspond to physiological and biochemical absorption features in vegetation. In general for Nitrogen (N) prediction, the important wavelength regions span the UV-Blue (360–450 nm), Red-Edge (600–700 nm), Near-Infrared (800–1400 nm), and Shortwave Infrared (1900–2500 nm). These nitrogen absorption features are consistent with previous findings of *Murray and Williams (1987)* [11], *Curran (1989)* [5], *Kokaly and Clark (1999)* [4] and *Curran et. al. (2001)* [12]. Among these nitrogen absorption features, the bands at 460 nm, 640 nm, and 660 nm correspond to chlorophyll absorption, which is strongly associated with nitrogen content. The remaining bands, except for 1645 nm, are linked to protein absorption, which is also closely related to nitrogen presence [13].

TABLE II: Important wavelength regions for Nitrogen (N) and Potassium (K) prediction using different spectral data.

	Raw (nm)	First Derivative	Continuum-Removed
N	369, 375, 1454 2273, 2274	376, 384, 450	624, 626, 636
		460, 670, 765	638, 681, 691
		798, 877, 918	1036, 1039, 1364
		1063, 1268, 1919	1377, 1439, 1441
		2081, 2087, 2268	1541, 1545, 1717
		2296, 2311, 2349	1720, 2062, 2069
		2458	2163
K	714, 1789, 2289	679, 935, 1704	662, 700, 928
		2046, 2180, 2232	871, 1265, 1287
		2266	

Similarly for Potassium (K) prediction, key wavelength regions include the Red-Edge and Near-Infrared (700–1300 nm) and Shortwave Infrared (1700–2300 nm). *Lu et. al. (2020)* [14] studied the sensitivity of different spectral bands and band indices to *Leaf Potassium Content (LKC)* in rice plant and showed that K sensitive bands are mainly located in short wave infrared region (1300 nm - 2500 nm) which they attribute to

TABLE I: Linear regression results for N and K using different spectral preprocessed data. S values are in absolute units and R^2 values are in percentage units. *The highlighted light green cells show best performance metric values obtained using each preprocessed data.*

Alpha (α)	Raw Spectra						First Derivative						Continuum Removed					
	N			K			N			K			N			K		
	S	Adj. R^2	5-fold- R^2	S	Adj. R^2	5-fold- R^2	S	Adj. R^2	5-fold- R^2	S	Adj. R^2	5-fold- R^2	S	Adj. R^2	5-fold- R^2	S	Adj. R^2	5-fold- R^2
0.005	0.14	43.37	41.11	0.16	12.12	11.39	0.05	91.77	89.97	0.11	57.83	54.87	0.15	35.19	32.17	0.15	18.76	15.46
0.01	0.14	43.37	41.11	0.16	12.12	11.39	0.04	96.09	94.20	0.10	64.75	60.94	0.14	44.91	39.63	0.14	31.34	28.70
0.05	0.13	48.86	46.01	0.15	22.21	18.56	x	x	x	x	x	x	0.09	78.18	67.78	0.13	39.95	38.03

the relationship of *LKC* with *LWC* (leaf water content). This outcome holds for most of our finding, except for few bands which are in VIS-NIR regions. However, it is interesting to see that almost all of the favorable bands explained in the literature for both N and K prediction are captured in the analysis using *first derivative spectra*, making it the best-performing approach compared to the other two.

Additionally, we evaluated model performance within the *Silicon wavelength range* (400 nm – 1100 nm) using same optimal α values determined above. Raw spectra failed to yield any meaningful outcome in this range ($\alpha = 0.05$). For the first derivative spectra ($\alpha = 0.01$), the model performed moderately well as before for both estimation. In N prediction, it yielded $S = 0.16$, adjusted $R^2 = 20.78\%$, and 5-fold $R^2 = 19.91\%$, with 441 nm and 915 nm identified as key wavelengths. In the case of K prediction, the model performed slightly better, yielding $S = 0.14$, adjusted $R^2 = 30.72\%$, and 5-fold $R^2 = 24.30\%$, with 472 nm, 676 nm, and 1088 nm entering the model. For the continuum-removed spectra ($\alpha = 0.05$), the model showed weak predictive capability for N estimation, with only 1046 nm being selected and resulting in poor performance ($S = 0.18$, adjusted $R^2 = 5.97\%$, 5-fold $R^2 = 3.86\%$). However, for K prediction, the model performed significantly better, achieving $S = 0.12$, adjusted $R^2 = 48.21\%$, and 5-fold $R^2 = 43.61\%$. Several wavelengths in the *Red-Edge and Near-Infrared (NIR) regions*: 700 nm, 728 nm, 871 nm, 1007 nm, 1023 nm, 1031 nm, and 1034 nm, were identified as important predictors. These selected wavelengths appear to be a *subset* of the key regions identified when using the full spectrum listed in Table I. The results also indicate that, while the first derivative and continuum-removed spectra can extract meaningful information within the Silicon range, their predictive power remains lower than when using the entire spectral range. Overall, these findings emphasize the significance of hyperspectral data in nutrient analysis and highlight the effectiveness of *spectral preprocessing* techniques and *algorithm tuning* in improving predictive accuracy.

IV. DISCUSSION AND CONCLUSION

This study exhibits the utility of hyperspectral data and spectral preprocessing in predicting foliar nitrogen (N) and potassium (K) content. Using stepwise linear regression we found that *first derivative spectra* achieves the highest accuracy at $\alpha = 0.01$ (N: 96.09%, K: 64.75%), underscoring its ability to amplify subtle spectral variations. The *continuum-removed spectra* also performed well, while the *raw spectra* yielded the lowest accuracy. Key predictive wavelengths range between regions- chlorophyll absorption (350–700 nm), red-edge reflectance (700–750 nm), near-infrared (800–1300 nm), and shortwave infrared (1900–2500 nm). A noticeable observation was the failure of the first derivative spectra at $\alpha = 0.05$, where too many predictor bands were selected, leading to overfitting. This shows the need of optimizing model by tuning the α -level. The results also emphasize the importance of preprocessing techniques in improving regression-based nutrient modeling. In conclusion, hyperspectral data, combined with preprocessing, appears to be a powerful tool for non-invasive nutrient monitoring. Future works can explore using

non-laboratory conditions data for such nutrient analysis using hyperspectral remote sensing techniques.

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